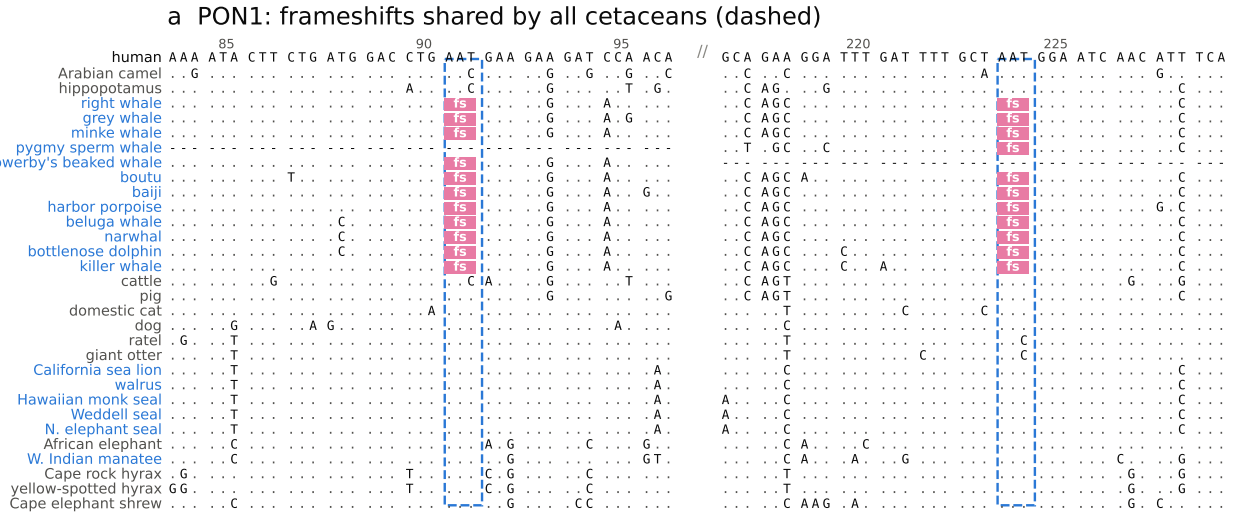
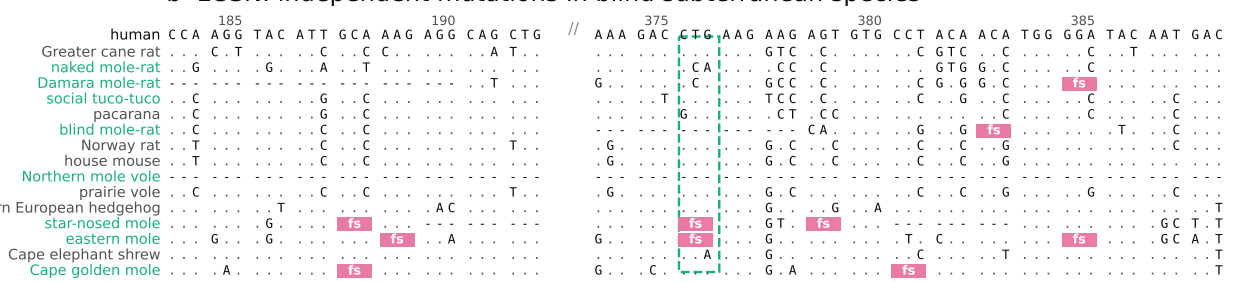
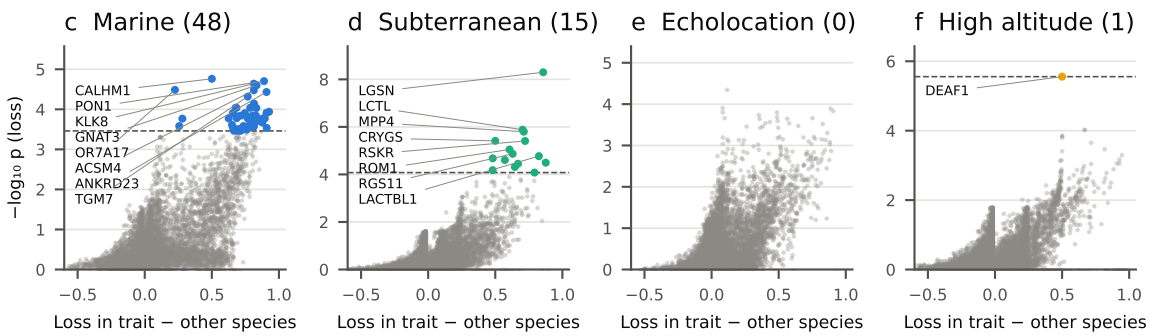






g Functions of the convergent losses

Marine (48 genes)

dots: lost in cetaceans, pinnipeds, manatee

Smell

OR7A17 ●●●	OR5M9 ●●●	OR2T1 ●●●	OR51G1 ●●●	OR5H15 ●●●
OR9Q2 ●●	OR5C1 ●●	OR5M3 ●●●	OR6C2 ●●●	OR51I1 ●●
OR51D1 ●●●	OR51I2 ●●	OR5H1 ●●●	OR51A7 ●●●	OR2K2 ●●●
OR5H14 ●●●	TAAR5 ●●●			

Taste

GNAT3 ●●	CALHM1 ●●	TAS1R3 ●●	TAS2R1 ●●●

Skin and hair

KRT1 ●●	KRT10 ●●	KRT79 ●●	KRTAP24-1 ●●	KLK8 ●●●
GSDMA ●●	SERPINB12 ●●	ANXA9 ●●		

Muscle

MYH6 ●●●	MYH13 ●●●	MYBPHL ●●●	MSS51 ●●●	ANKRD23 ●●●

Detoxification and metabolism

PON1 ●●●	ACSM4 ●●●	GLYATL3 ●●●	SULT6B1 ●●●	PTGR1 ●●
GBA3 ●●●				

Other

SLC4A9 ●●	ATP13A5 ●●●	TGM7 ●●●	HTRA4 ●●●	MRGPRX2 ●●
ADGRF2 ●●	CCDC170 ●●	TLE7 ● ●		

Subterranean (15 genes)

Lens

LGSN ● ●●●	LCTL ● ●●●
CRYGS ● ● ●	CRYBA4 ● ●●●
BFSP1 ● ● ●	GJA8 ● ● ●

Retina

MPP4 ●●●●●	ROM1 ● ●●●
RGS11 ● ●●●	CABP5 ● ●●●

Vitreous

OPTC ●● ●●●	

Other

RSKR ● ●●●	LACTBL1 ● ●●●
B3GNT4 ●● ●●●	TPD52L3 ●●●●●●

dots: lost in each of the 6 origins

Black outline: loss reported in earlier studies